

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INFORMATION TECHNOLOGY

Paper 032 – General Proficiency

ALTERNATIVE TO SCHOOL-BASED ASSESSMENT

*1 hour 30 minutes***READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Code is to be written in Pascal.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2015 Caribbean Examinations Council
All rights reserved.



Answer ALL questions.

1. WORD PROCESSING

Consider the letter shown below.

GY Bank Ltd
Regent Street, St Linden
Tel: 599-212-9874; Website: www.gb.com

May 23, 2015
Dr Bud Lee
50 Main Street
Jamestown

Dear Dr Lee

With effect from July 1, 2015, we will be introducing our online banking service. This new service will enable you to do the following:

- Access you account online via the Internet
- Pay your utility bills online
- Transfer funds between accounts

Please visit our bank and apply for an online account. In addition, please check our website www.gb.com for more information on our new services.

Yours sincerely

Carol Smith
Branch Manager



- (a) State THREE formatting features used on the name of the bank.

.....

.....

.....

(3 marks)

- (b) List THREE steps required to insert a logo located on a flash drive to the right of the name and address of the bank.

.....

.....

.....

(3 marks)

- (c) Identify ONE position within the letter where word wrap was used.

.....

.....

(1 mark)

- (d) The secretary wishes to send this letter to other customers with their names and addresses included in the letter using the mail merge feature. Name TWO documents which are required to perform the mail merge.

.....

.....

(2 marks)

- (e) Identify THREE merge fields that this letter should have.

.....

.....

.....

(3 marks)



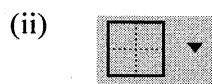
(f) State the use of EACH of the following icons in a word processing program:



.....

.....

(2 marks)



.....

.....

(2 marks)

(g) The secretary wishes to change ALL occurrences of 2015 to 2016. Which feature would achieve this?

.....

(2 marks)

(h) The spellcheck feature did not flag 'Access you account online' in the letter. Explain why this did not happen.

.....

.....

(1 mark)

(i) A word processing feature highlighted the phrase *gb.com* in your letter. Explain why this phrase would have been highlighted.

.....

.....

(1 mark)

Total 20 marks



2. SPREADSHEET

The amount of money spent to support the Community Policing Programme in the Divisions of a country for the past three years is given in the worksheet below.

	A	B	C	D	E	F	G
1	COMMUNITY POLICING SUPPORT PROGRAMME						
2							
3	DIVISION	2012	2013	2014	TOTAL	%ofTotal	
4	A	1,300	1,500	2,500	5,300	40.77%	
5	B	700	900	1,100	2,700	20.77%	
6	C	400	500	900	1,800	13.85%	
7	D	200	600	1,200	2,000	15.38%	
8	E	100	300	800	1,200	9.23%	
9	TOTAL	2,700	3,800	6,500	13,000		
10							

- (a) State the range of cells which contains the total amount of money spent *each year* to support community policing.

(1 mark)

- (b) What text alignment has been applied to the heading of columns with numeric data?

(1 mark)

- (c) The formula =E4/\$E\$9 was entered in cell F4 and copied to cells F5 to F8.

- (i) State the type of cell addressing used in the numerator (E4) and in the denominator (\$E\$9).

(2 marks)

- (ii) What is the reason for using \$E\$9?

(1 mark)



- (d) What data format has been applied to the range (F4..F8)?

.....
(1 mark)

- (e) Write a FUNCTION to find the *average of the totals* from 2012 to 2014.

.....
(2 marks)

- (f) Complete the criteria range below to filter the data for all divisions that received more than \$500 in 2012.

DIVISION	2012	2013	2014

(1 mark)

- (g) State the divisions that have met the criterion from Part (f).

.....
(2 marks)

- (h) The title in row 1 is centred across the columns used to create the worksheet. Which feature of the spreadsheet program has been used to achieve this?

.....
(1 mark)

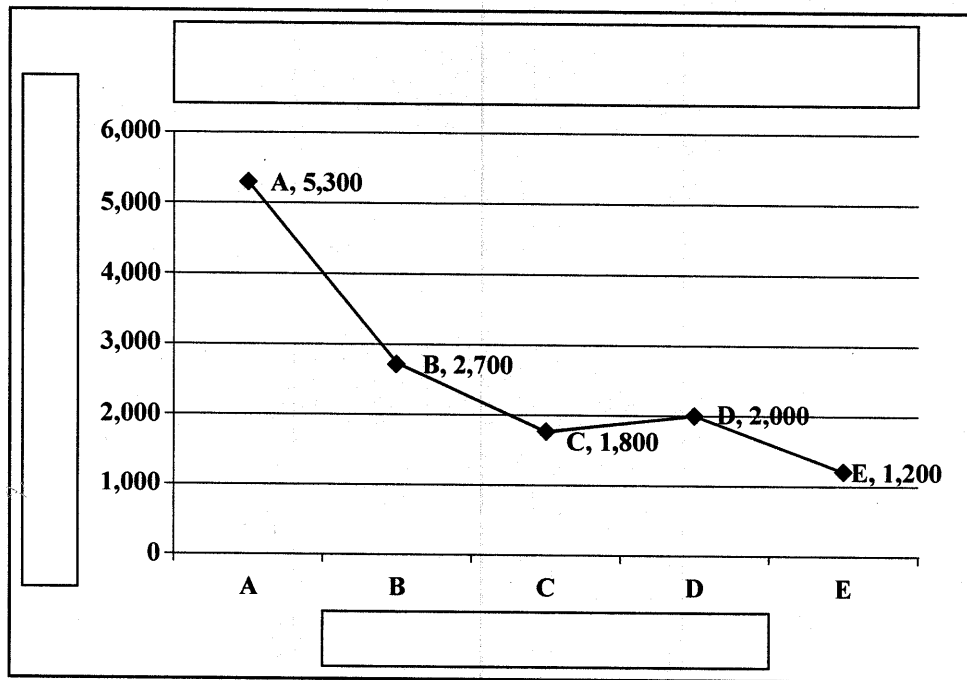
- (i) The data in the worksheet has been sorted in ascending order by the %ofTotal column. Which division is now at the top of the sorted worksheet?

.....
(1 mark)

GO ON TO THE NEXT PAGE



- (j) The following chart was generated from the data in the worksheet.



- (i) State the type of chart shown.

(1 mark)

- (ii) Add appropriate labels for the **x-axis**, **y-axis** and the **title** in the boxes provided in the chart.

(4 marks)

- (iii) State the category range and the data range used to create the chart.

(2 marks)

Total 20 marks



3. DATABASE MANAGEMENT

Consider the database CRIME with two tables shown below.

crime_type	
Ccode	Category
CM1	Murder
CM2	Robbery
CM3	Domestic Violence

crime_stats		
Ccode	Division	Committed
CM3	A	50
CM3	C	80
CM1	A	30
CM1	E	10
CM2	B	120
CM2	D	75
CM2	A	200
CM1	B	20
CM3	D	15
CM2	C	90
CM3	E	20

- (a) State the data type of the **Committed** and **Category** fields.

.....

 (2 marks)

- (b) How many records and fields are there in the **crime_type** table?

.....

 (2 marks)



- (c) In the **crime_stats** table, why is the **Ccode** field unsuitable as a primary key?

.....

.....

(1 mark)

- (d) If the tables are joined by the **Ccode** fields, how many robberies are committed in Division B?

.....

(1 mark)

- (e) The result of a query is given below.

Query1	
SumOfCommitted	
	710

- (i) Which of the two tables is used to perform the query?

.....

(1 mark)

- (ii) What is the purpose of the query?

.....

(1 mark)

- (f) It is expected that crimes committed in 2015 will be reduced by 5% compared with the 2014 figures. Write the formula for a calculated field to show the expected crime for 2015. Name this field **target2015**.

.....

(2 marks)



- (g) The following query was created from the data in the database.

Field:	Category	Division	Committed
Table:	crime_type	crime_stats	crime_stats
Sort:			
Show:	☒	☒	☒
Criteria:	"Domestic Violence"		>50
or:			

- (i) What is the purpose of the query?

.....

.....

(3 marks)

- (ii) How many records will be displayed when the query is executed?

.....

(1 mark)



- (h) The following report was generated from the data in the database.

Crime Statistics for 2014

CCode	Division	Committed
CM1	A	30
	B	20
	E	10
	Sum	60
CM2	A	200
	B	120
	C	90
	D	75
	Sum	485
CM3	A	50
	C	80
	D	15
	E	20
	Sum	165
Grand Total		710

State:

- (i) The table used to create the report (1 mark)
- (ii) The title of the report (1 mark)
- (iii) The field used to group the records (1 mark)
- (iv) Another field that can be used to group the records (1 mark)
- (v) What 710 represents in the report (1 mark)
- (vi) The feature that was used to produce the figure in (v) (1 mark)

Total 20 marks

GO ON TO THE NEXT PAGE



4. PROBLEMSOLVING AND PROGRAMMING

Consider the following program written in Pascal.

Line No	Code
1	Program test;
2	Var
3	score : array[1..6] of integer;
4	mark, count : integer;
5	Begin
6	mark := score[1];
7	For count := 2 to 6 DO
8	Begin
9	IF mark > score[count] THEN
10	mark := score[count];
11	End;
12	Writeln(mark);
13	{print the result of the program}
14	End.

- (a) For the array declared in the program, state the
- (i) name of the array
 - (ii) size of the array
 - (iii) type of data that will be stored in the array.
- (3 marks)
- (b) For EACH of the following, write "YES" and the line number if the operation is found in the program above, otherwise write "NO".
- (i) A loop
 - (ii) An arithmetic statement
 - (iii) A conditional statement
 - (iv) A logic operation
- (7 marks)

GO ON TO THE NEXT PAGE



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (c) Explain the purpose of the code in line 6.

.....
.....

(2 marks)

- (d) State the line number of an output statement.

.....

(1 mark)

- (e) Write the Pascal code to store a mark of 13 in the last element of the array.

.....
.....

(3 marks)

- (f) (i) What type of error would occur if the semicolon (;) was missing at the end of line 1?

.....

- (ii) When would this error be detected?

.....
.....

(2 marks)

- (g) What type of error would occur if the program did not produce the correct results?

.....

(1 mark)

- (h) Indicate the line number where a comment is used.

.....

(1 mark)



- (c) Explain the purpose of the code in line 6.

.....
.....
(2 marks)

- (d) State the line number of an output statement.

.....
(1 mark)

- (e) Write the Pascal code to store a mark of 13 in the last element of the array.

.....
.....
(3 marks)

- (f) (i) What type of error would occur if the semicolon (;) was missing at the end of line 1?

.....

- (ii) When would this error be detected?

.....
.....
(2 marks)

- (g) What type of error would occur if the program did not produce the correct results?

.....
(1 mark)

- (h) Indicate the line number where a comment is used.

.....
(1 mark)

GO ON TO THE NEXT PAGE

01229032/JANUARY/F 2016



0122903215

- (i) Consider the following program written in Pascal.

Line No	Code
1	Program test;
2	Var
3	score : array[1..6] of integer;
4	mark, count : integer;
5	Begin
6	mark := score[1];
7	For count := 2 to 6 DO
8	Begin
9	IF mark > score[count] THEN
10	mark := score[count];
11	End;
12	Writeln(mark);
13	{print the result of the program}
14	End.

Complete the following trace table using the values (18, 20, 7, 9, 5, 13) that were entered when the program was executed.

Count	Score[count]	Mark
1	18	18
2	20	
3	7	
4	9	
5	5	
6	13	

(5 marks)

- (j) What is the purpose of the program?

.....

.....

(1 mark)

GO ON TO THE NEXT PAGE

